

ChimeraBoost: Gradient Boosted Oblivious Trees in Pure Python

Version 0.33.0 · September 2026 · Apache-2.0 · github.com/bbstats/chimeraboost

Abstract

Gradient boosted decision trees remain the standard method for prediction on tabular data, and the leading implementations are large C++ systems. This paper describes ChimeraBoost, a gradient boosting library implemented entirely in Python on four dependencies (numpy, numba, scipy, and scikit-learn), with no compiled extension and no GPU requirement. On TabArena, a community benchmark run by its own maintainers and treated here strictly as a held-out evaluation, ChimeraBoost's default configuration ranks second among gradient boosting libraries, behind CatBoost and well ahead of XGBoost and LightGBM, while training in under half of CatBoost's median time. Much of the speed comes from a structure-replay refit: after early stopping, the selected tree structures are replayed against gradients computed on all rows, recovering the withheld validation data at negligible cost and reducing total fit time by 34.8% on a 59-dataset suite. Configuration choices that are usually left to the user, such as whether leaves hold constants or linear models, are made during the fit by racing candidate configurations on validation data under a shared round budget. Beyond the point prediction that TabArena measures, the library also estimates full predictive distributions: a gradient-matrix-free multi-quantile head fits a grid of conditional quantiles in one booster, with predictions that provably never cross, at approximately the per-round cost of a single level. Exact TreeSHAP attributions, conformalized prediction intervals, and temperature-calibrated probabilities are provided without external packages.

1. Introduction

The dominant gradient boosting implementations, XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018), owe their speed to substantial C++ codebases. That design choice makes them hard to modify: testing a new algorithmic idea means writing C++ inside a large codebase, which few practitioners can do. ChimeraBoost explores the opposite corner of the design space. The entire library is Python, roughly 11,700 lines across 13 modules, with numba supplying JIT-compiled kernels where profiling shows they are needed. The question is whether a library built this way can be competitive with the compiled systems rather than merely convenient, and the evidence presented below suggests that it can.

The contributions are as follows:

1. **A competitive pure-Python GBDT.** On the TabArena leaderboard the default configuration scores Elo 1297, second among gradient boosting libraries after CatBoost (1357) and ahead of XGBoost (1208) and LightGBM (1181), at a median training time of 2.65 seconds per thousand rows against CatBoost's 5.88 (section 7).
2. **Structure-replay refit.** Early stopping withholds a validation fold, so the selected model has seen only part of the data. ChimeraBoost replays the selected tree structures against gradients computed on all rows, refitting leaf values only. This removed a step that had consumed 37 to 49% of every default fit and reduced total fit time by 34.8% on the decision suite, with accuracy unchanged (section 4).
3. **A gradient-matrix-free multi-quantile head.** A single booster estimates a grid of conditional quantiles (19 levels by default) with one tree structure per round and a vector-valued leaf. A projection collapses each row's K-channel pinball gradient to a scalar in one pass, so the $(n \times K)$ gradient matrix is never materialized and the head costs approximately what a single level costs per round. Predictions are monotone across levels by construction (section 6).
4. **An evaluation protocol with strict suite separation.** Development, tuning, published evidence, and external validation use disjoint dataset collections, and the external leaderboard is never consulted when making changes (section 7).

2. System overview

ChimeraBoost exposes a scikit-learn-compatible API of seven public names, three of which are estimators: `ChimeraBoostRegressor`, `ChimeraBoostClassifier`, and `ChimeraBoostQuantileRegressor`. Supported objectives include squared error, absolute error, Huber, Poisson, Gamma, Tweedie, and user-defined objectives for regression; binary and multiclass log loss for classification; and joint multi-quantile estimation.

Data handling. Categorical features are processed natively with ordered target statistics in the manner of CatBoost, averaged over four permutations, and may be addressed by column index or by DataFrame column name. Missing values receive a dedicated histogram bin at both fit and predict time, for numeric and categorical features alike. No one-hot encoding, imputation, or label encoding is required, and pandas is intentionally not a dependency: DataFrames are consumed through their own `to_numpy` method.

Automatic configuration. Several decisions usually delegated to the user are made during the fit. Linear leaf models (Shi, Li and Li, 2019) are auditioned against constant leaves, and automatically generated cross features (Zhang et al., 2023) against plain features, in short races on a shared round budget; section 5 details the mechanism. The learning rate adapts to training set size; CatBoost's defaults do the same, and dataset size is the only quantity they vary with. Early stopping is enabled by default, and the stopped model is refit on all rows by structure replay (section 4). The remaining speed-

accuracy choice is a single integer, `quality`, whose five settings were each measured to sit on the speed-accuracy frontier:

quality	profile	fit time (default = 1x)
1	fast	0.4x
2	balanced	0.8x
3	accurate (default)	1x
4	ensemble of 5	2.6x
5	ensemble of 8	3.8x

Interpretability and calibration. The `shap_values` method computes exact interventional TreeSHAP attributions (Lundberg et al., 2020) with no external package; the computation is exact over its background set, which defaults to at most 200 rows, and costs roughly 3 ms per row at depth 6. Classifier probabilities are temperature-scaled on the early-stopping fold; the transformation is monotone, leaving accuracy and AUC unchanged while improving calibration.

3. Training pipeline design

Oblivious trees. Every node at a given depth shares one (feature, threshold) pair, so a tree of depth d is described by d splits and a leaf index is a d -bit number (Kohavi and Li, 1995; Prokhorenkova et al., 2018). Prediction reduces to a handful of comparisons and a table lookup per tree, executed in a single numba kernel parallelized over samples so that each row traverses the entire forest while its features remain in cache. Sharing one split across a depth also acts as a strong regularizer.

Quantized histogram split finding. Splits are searched over binned features (128 bins by default), with gradients and Hessians quantized to 15-bit integers following Shi, Ke et al. (2022). Integer histograms reduced fit time by 20 to 25% at unchanged accuracy in our measurements. Leaf values are always computed from exact floating-point gradients, and results are deterministic for a fixed seed. When row subsampling is enabled, rows are drawn by minimum-variance sampling (Ibragimov and Gusev, 2019), which weights selection by gradient magnitude.

Vector leaves. Multiclass models grow one tree per round rather than one per class, storing a class vector in each leaf (Iosipoi and Vakhrushev, 2022; Zhang and Jung, 2021). This made multiclass fits 2.5 times faster while improving Brier score by 5% at equal F1. The same mechanism supplies the quantile head's per-leaf vector.

Bagging without replacement. The `n_ensembles` option fits members on 80% subsamples drawn without replacement, which outperformed the classical bootstrap on both accuracy and fit time in our comparisons. Members fit in parallel worker processes sharing one thread budget. With

`refit_members=True`, each member is replayed on its full subsample (section 4); five refit members outperformed eight plain members on both accuracy and speed.

4. Structure-replay refit

Choosing the tree count by early stopping requires a validation fold, so the selected model has never trained on those rows, and on small datasets the withheld rows are a substantial share of the training data. The standard fix, adopted by AutoGluon's `refit_full` (Erickson et al., 2020) among others, retrains from scratch on all rows once the count is known. The retrain recovers the accuracy but costs a second fit: profiled on ChimeraBoost's own pipeline, it consumed 37 to 49% of every default fit, the single largest component of fit time.

Most of that second fit repeats work the first fit already did. Growing trees, the histogram construction and split search, accounts for 83 to 85% of a fit, and a from-scratch retrain spends that time rediscovering split structures the first model already found. An oblivious tree's structure is also compact: a depth- d tree is d (feature, threshold) pairs, independent of its leaf values. ChimeraBoost therefore replays rather than retrains. The recorded structures are applied round by round to gradients computed on the full data, and only the leaf values are refit by the usual Newton step, so the gradient sequence evolves exactly as boosting requires while the split search is never repeated. The held-out rows now contribute to the leaf estimates, the same benefit the full retrain provided, and the whole procedure costs approximately the leaf-estimation share of a fit. Both halves of the idea exist separately in prior systems: refreshing leaf values on a fixed structure is XGBoost's `refresh` updater and LightGBM's `refit()`, and the full-data retrain after early stopping is established practice. Combining the two, and enabling the combination by default, is the contribution here.

One implementation detail deserves mention because getting it wrong produces convincing but invalid results. Split thresholds are bin indices, meaningful only under the binner that produced them, so the donor model's preprocessor must be reused verbatim. The first implementation refit the binner on the full data, which silently moved every threshold underneath the replayed structures; the apparent accuracy gains of that version were target leakage, and the project's leakage regression test caught it. The results below are from the corrected version.

Measured on the 59-dataset decision suite, replay cut total default fit time by 34.8%, with 58 of 59 datasets fitting faster; on the high-cardinality suite the saving was 15.2% (faster on 12 of 14), smaller because replay does not cover multiclass and categorical preprocessing is a fixed cost either way. Accuracy was statistically unchanged on four independent suites, with every median difference essentially zero (+0.005%, -0.017%, -0.043%, -0.114%), two of which played no part in the ship decision. The same replay serves the bagged ensembles: `refit_members` replays each member on its full subsample, recovering the rows that member held out for out-of-bag estimation.

5. Selecting the configuration during the fit

Several of ChimeraBoost's defaults are decisions rather than fixed values, because the optimal configuration depends on the dataset. Whether leaves should hold ridge linear models (Shi, Li and Li, 2019) or constants depends on how much within-leaf structure the data has; whether generated cross features (Zhang et al., 2023) improve the model depends on how interaction-heavy the data is. Standard practice delegates such choices to user-side cross-validation. ChimeraBoost instead treats each option as a candidate and races the candidates inside the fit, in the manner of Hoeffding races (Maron and Moore, 1993) and successive halving (Jamieson and Talwalkar, 2016).

The mechanism is simple. Each candidate configuration trains on the same train/validation split for a shared budget of `selection_rounds` boosting rounds (100 by default), candidates are judged on their best validation loss within the budget, and only the winner continues to full early stopping; a candidate that early-stops inside the budget has already completed its full fit. Racing makes fits about 1.5 times faster than training every candidate to completion, and it works together with the replay of section 4: the winner, once early-stopped, is replayed on all rows, so a configuration selected on partial data still produces a full-data model.

Two races run under this budget. The regressor auditions linear leaves against constant leaves (binary classification enables linear leaves directly, and below roughly 1,000 rows the audition is skipped in favor of constants). The second race refits with generated cross features, differences and products of the top-ranked numeric pairs plus group-centered columns of top numerics against top categoricals, and keeps whichever model validates better. Each decision is exposed afterward as a fitted attribute (`linear_leaves_selected_`, `cross_features_selected_`), so the selected configuration can be inspected after fitting.

We empirically verified that shortening the budget degrades the selection. A 25-round race was evaluated and rejected: validation loss that early in a fit misranks candidates often enough to lose accuracy, and the full-data refit then amplifies the wrong pick rather than repairing it. Substituting a different selection statistic at 25 rounds lost accuracy as well, indicating that the loss came from the shortened budget rather than from the selection rule. At 100 rounds the audition's choices agree with full runs closely enough that the disagreements are within noise, and users who want to skip the races entirely can pin both decisions with `quality=1`, which is how that setting reaches its 0.4x fit time.

6. The multi-quantile head

`ChimeraBoostQuantileRegressor` estimates a grid of conditional quantiles jointly. From one fitted grid the model answers several distributional queries at no additional cost: central intervals, the median, the mean, the CDF at arbitrary thresholds, inverse-transform samples, and per-row exceedance probabilities via `predict_thresh(X, t)`.

Split search. Scoring K pinball objectives per candidate split would multiply the histogram work by K . Instead, a projection collapses each row's K -channel gradient to a scalar in a single pass, and split gains computed on the projected gradient land within 7% of the exact summed gain. Not every projection works: a naive sum across a symmetric grid is blind to dispersion, because the gradient contributions of levels τ and $1 - \tau$ cancel exactly for an interval that is correctly centered but too narrow.

Non-crossing guarantee. Each row's predicted quantiles are sorted on output. Rearrangement never increases pinball loss at any level (Chernozhukov, Fernández-Val and Galichon, 2010), so the guarantee is free. The property is absent from per-level ensembles in practice: across 36 real regression datasets, LightGBM per-level boosters reversed 22% of adjacent level pairs on average and crossed on every dataset, and CatBoost's shared MultiQuantile head also crossed on every dataset. ChimeraBoost's crossing rate was exactly zero on all 36.

Calibration. The raw grid's nominal 80% and 90% intervals delivered 77% and 87% observed coverage, closer to nominal than LightGBM per-level (72% and 83%) or CatBoost MultiQuantile (73% and 83%). With `conformalize=True` the intervals gain distribution-free marginal coverage by conformalized quantile regression (Romano, Patterson and Candès, 2019), measured within 2.7 percentage points of nominal at $n = 10,000$. An out-of-fold study on seven datasets confirmed the derived quantities: exceedance probabilities showed expected calibration error from 0.029 to 0.051, crossing remained zero everywhere, and the raw grid's slight narrowness was confined to the outermost tail cells. Conformalization corrected the tail cells specifically, bringing 90% coverage to between 0.902 and 0.905 on the three worst datasets, at a cost of about one point of CRPS skill.

Attribution of interval width. Because both edges of an interval come from one model, SHAP attributions can be computed for the width of an interval, identifying the features that make a given row's prediction uncertain rather than high or low. A stack of independently fitted per-level models cannot produce this decomposition.

Comparison. On the 36 datasets, with every method early-stopping on the same split: against 19 independent LightGBM quantile boosters, CRPS was approximately tied (23 wins, 13 losses), while the interval score, a proper scoring rule that accounts for coverage and width jointly, favored ChimeraBoost 31 to 5 at two thirds of the fit time. CatBoost's MultiQuantile won on CRPS; section 8 quantifies this.

7. Evaluation

Protocol. Benchmark suites are assigned single, non-overlapping roles: a synthetic screen for mechanism checks; a decision suite (Grinsztajn et al., 2022, plus a frozen high-cardinality collection) on which changes are accepted or rejected by per-stratum sign tests; a tuning suite (PMLB), the only data hyperparameters are fitted to; a 22-dataset independently audited public suite for published evidence; and TabArena, which is reported and never consulted during development. A suite used for tuning cannot also measure overfitting to itself, and datasets enter a suite on data properties alone (row counts, cardinality, missingness), never on observed results. When early speed claims for the quantile head (3

to 6 times LightGBM) were traced to fixed-round fits on synthetic data, the documentation was corrected to the real-data figure of 1.5x, with the correction noted in place.

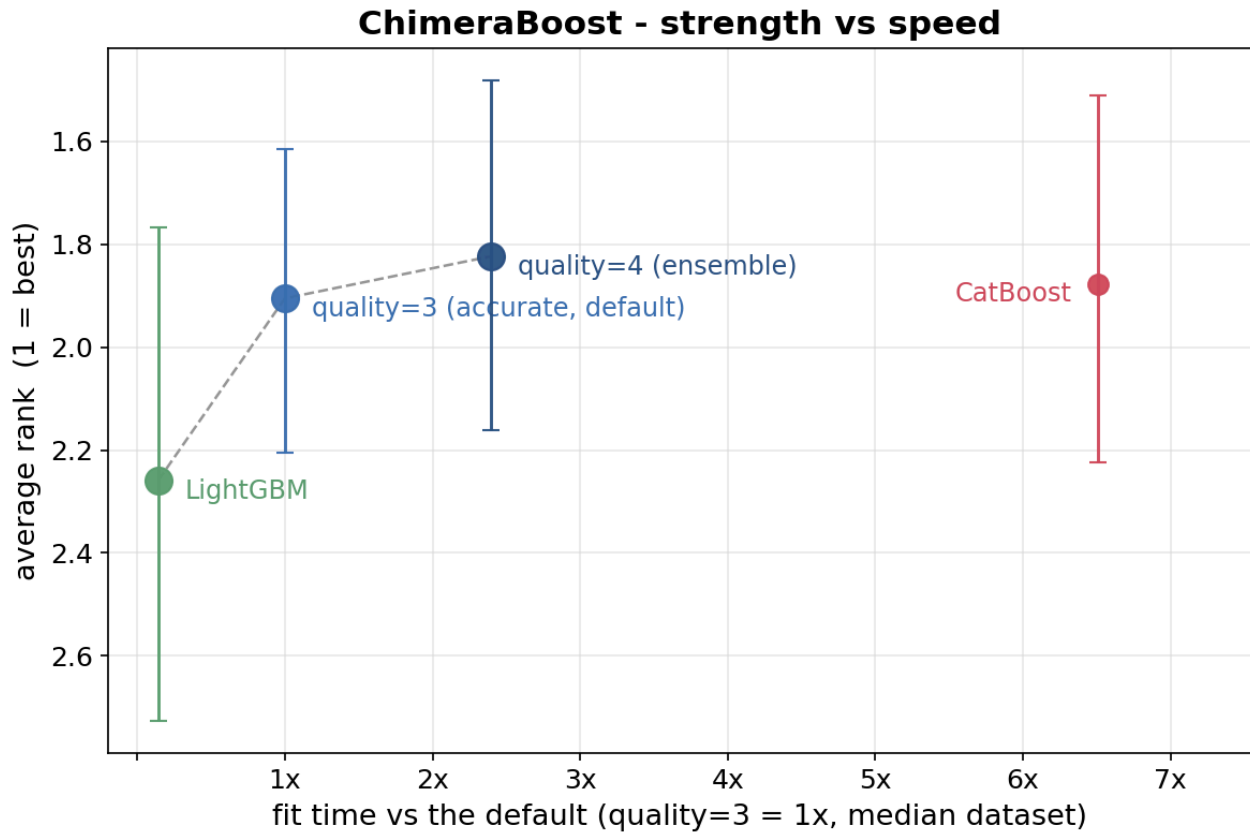
TabArena. TabArena (Erickson et al., 2025) is a leaderboard of 51 curated classification and regression datasets in which every model is run by the benchmark's maintainers and ranked by Elo. As of September 2026, in the leaderboard's default view, gradient boosting libraries at default settings ranked as follows:

model (default config)	Elo	median train s/1K	median predict s/1K
CatBoost	1357	5.88	0.025
ChimeraBoost	1297	2.65	0.052
XGBoost	1208	1.94	0.123
EBM	1191	6.67	0.014
LightGBM	1181	1.96	0.142

ChimeraBoost's default placed second, with the 89-point gap to third place exceeding the 60-point gap to first. It trained in 45% of CatBoost's median time per thousand rows and predicted two to three times faster than XGBoost and LightGBM defaults, though CatBoost's predict time remains lower. The tuned-and-ensembled entry scored 1360, marginally above CatBoost's default, using a search space that predates several of the library's current adaptive defaults. The top of the overall board is held by tabular foundation models near Elo 1750, pretrained networks that use far more compute than any GBDT. The evaluated entry ran version 0.30.0; the two releases since left default-configuration predictions unchanged bit for bit, so the accuracy figures carry over, and later speedups mean the timing column, if anything, understates the current version. Since no TabArena result has ever influenced a change to the library, these figures constitute an out-of-sample evaluation.

Public suite. The published chart runs on 22 audited public datasets disjoint from all development data. In the most recent full run (August 2026, three seeds, 21 of 22 datasets scored), with fit times expressed relative to the ChimeraBoost default:

model	avg rank	95% CI	median fit time (default = 1x)
ChimeraBoost quality=4 (ensemble)	1.82	1.48-2.16	2.4x
CatBoost	1.88	1.51-2.22	6.5x
ChimeraBoost default	1.90	1.62-2.21	1.0x
LightGBM	2.26	1.77-2.73	0.1x



Each ChimeraBoost configuration is ranked in its own three-way contest against CatBoost and LightGBM, so 2.0 is the middle of the field. Ranks are facet-balanced across task type, dataset size, and cardinality; the unweighted ranks, which favor CatBoost slightly, are recorded in the repository. The default configuration was statistically indistinguishable from CatBoost at about a sixth of CatBoost's fit time on the median dataset, and the `quality=4` ensemble achieved the best average rank at just over a third of CatBoost's fit time.

8. Limitations

CRPS against CatBoost MultiQuantile. CatBoost's shared quantile head won 29 of 36 datasets on CRPS, a real deficit. It required a median 8.3 times ChimeraBoost's fit time and was substantially worse calibrated (worst coverage error 0.64 against a nominal 0.90, versus 0.10 for ChimeraBoost), so the interval score favored ChimeraBoost 25 to 11; applications that weight CRPS alone and disregard cost are better served by CatBoost.

Compiler cold start. The first fit after a fresh install or upgrade incurs roughly ten seconds of numba compilation (0.8 s warm, 0.27 s in steady state). A `chimeraboost-warmup` command and an environment variable allow precompilation; the cost is reduced rather than eliminated.

Training speed, especially at scale. LightGBM and XGBoost defaults train in roughly 75% of ChimeraBoost's time at TabArena's scale, and the gap grows with row count: in internal measurements on synthetic data at a fixed tree count, the fit-time ratio to LightGBM widened from about 5x at 50,000 rows to roughly 10x at 500,000, because numba's generated histogram kernels do not match hand-tuned C++ SIMD. The compiled libraries offer GPU training for exactly these larger datasets; the pure-Python design rules out a GPU backend, so ChimeraBoost has no equivalent at that scale. The fit-time comparisons in this paper were measured on datasets well under a million rows and should not be extrapolated past that. Prediction throughput is less affected; at two million rows it measured on par with LightGBM. ChimeraBoost targets quality per unit of compute rather than minimum training time.

API stability. The library is at version 0.x; names have been stable in practice but are not yet guaranteed.

9. Availability

ChimeraBoost is available on PyPI (`pip install chimeraboost`, Python 3.9+) under the Apache-2.0 license.

```
from chimeraboost import ChimeraBoostClassifier, ChimeraBoostQuantileRegressor

clf = ChimeraBoostClassifier()                # or quality=4 for the ensemble
clf.fit(X_train, y_train, cat_features=["city"]) # categoricals by name, NaNs fine
proba = clf.predict_proba(X_test)             # temperature-calibrated

qr = ChimeraBoostQuantileRegressor(conformalize=True)
qr.fit(X_train, y_num, cat_features=["city"])
lo, hi = qr.predict(X_test, kind="interval", alpha=0.1).T # guaranteed 90% band
p_over = qr.predict_thresh(X_test, 100.0)             # P(y > 100) per row
```

Documentation is at bbstats.github.io/chimeraboost. The benchmark harness, raw results, and the studies cited here are in the [repository](#) under `benchmarks/`: the TabArena figures are the leaderboard read of 2026-09-01 (`benchmarks/tabarena/LEADERBOARD_SNAPSHOT.md`), the public-suite run is recorded in `benchmarks/PUBLIC_PLAN.md`, the quantile comparisons come from `benchmarks/quantile_suite.py` ([full tables](#)), and the out-of-fold calibration study is `benchmarks/quantile_oof_calibration.py`. The large-data scaling measurements come from `benchmarks/scaling_giant.py` and `benchmarks/scaling_predict.py`. Per-feature attributions for borrowed techniques are maintained on [the attribution page](#).

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